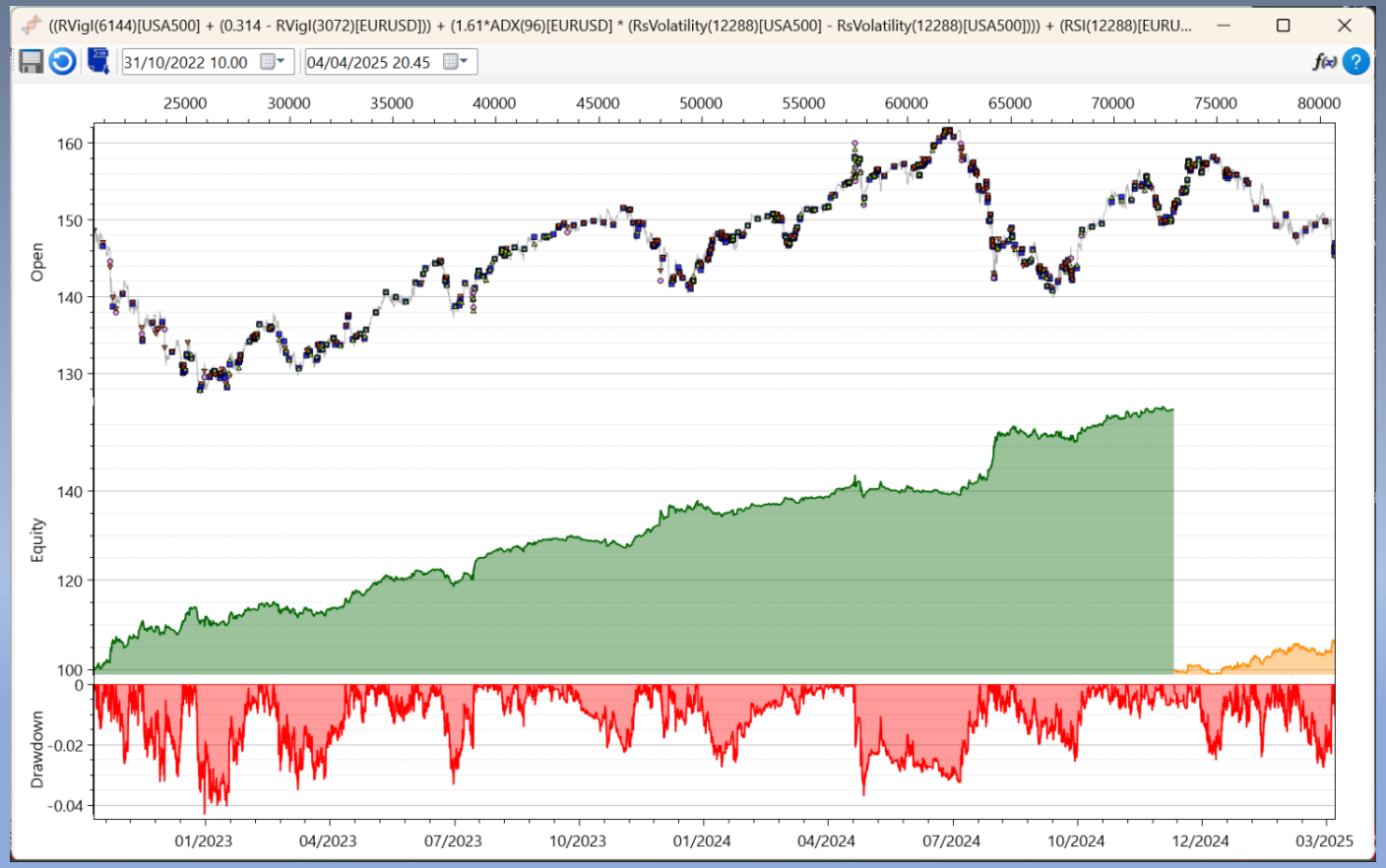
Discovering & Optimizing Trading Strategies with

Sharpshooter

Genetic Programming Beyond Overfitting



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Version 10/2026 by Lauri Suoranta

Background

Genetic programming has been applied in systematic trading for two decades:

- Well-known, commercially available software called StrategyQuant has been selling for over a decade. Oriented towards retail traders with substantial amount of trading capital.
- Traders with StrategyQuant probably made good profits with it in 2010s.[1] Continued development suggests that quite a few continue to see value in it. Whether this is justified is an open question.

[1] We return to the reasoning behind this claim later on.

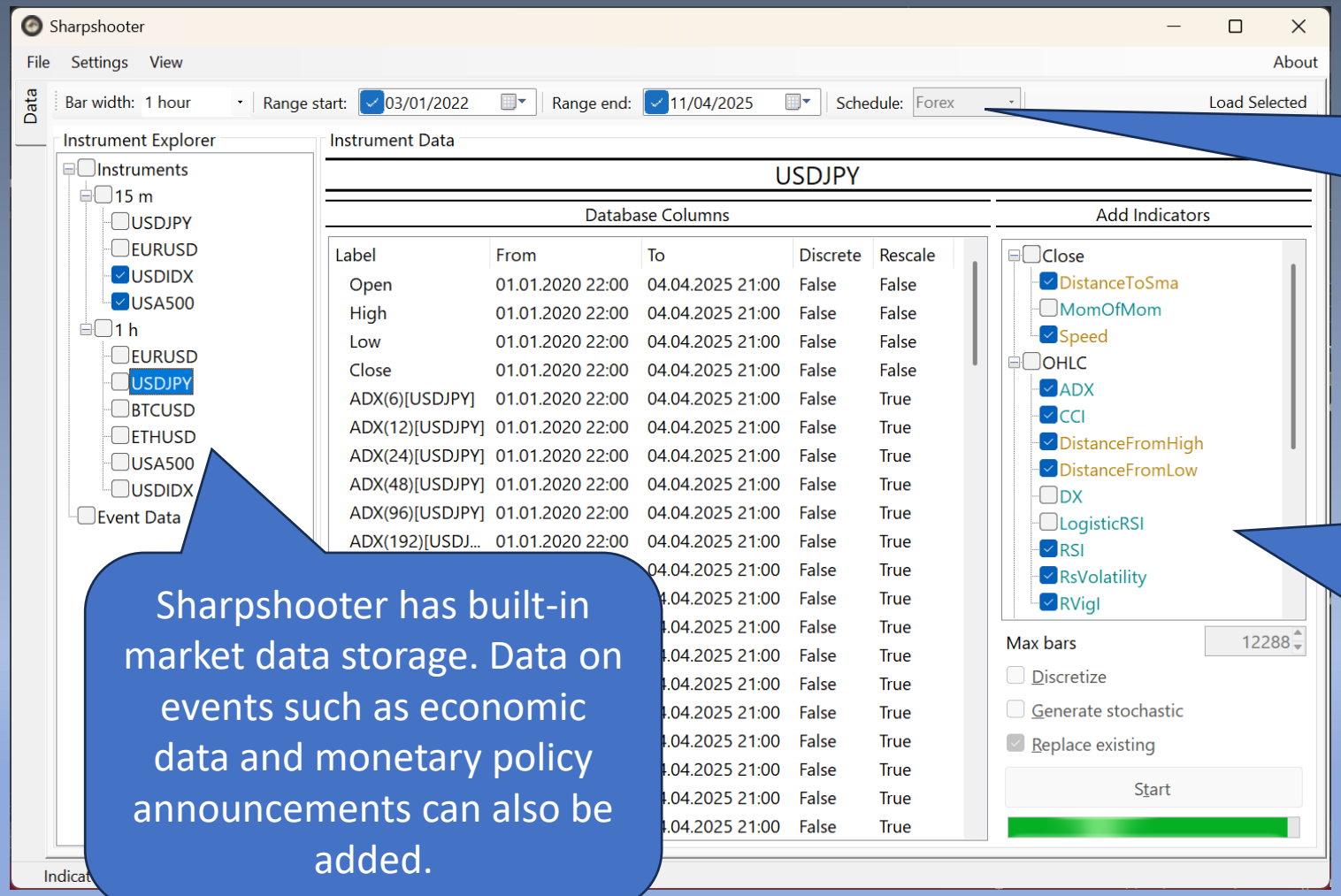
Tour of Sharpshooter — Introduction

Sharpshooter is an extensible, primarily GP-based trading strategy discovery engine/software. Oriented towards FX trading.

Has a fair amount of similarity to StrategyQuant, but oriented strongly towards institutional use in its features:

- More powerful extensibility features that aligned institutional workflows: extended with standard software engineering tools (Microsoft Visual Studio) rather than within the software itself.
- Programmatic access allows integration into other proprietary software systems.
- Highly advanced strategy execution agent.

Tour of Sharpshooter — Data Management

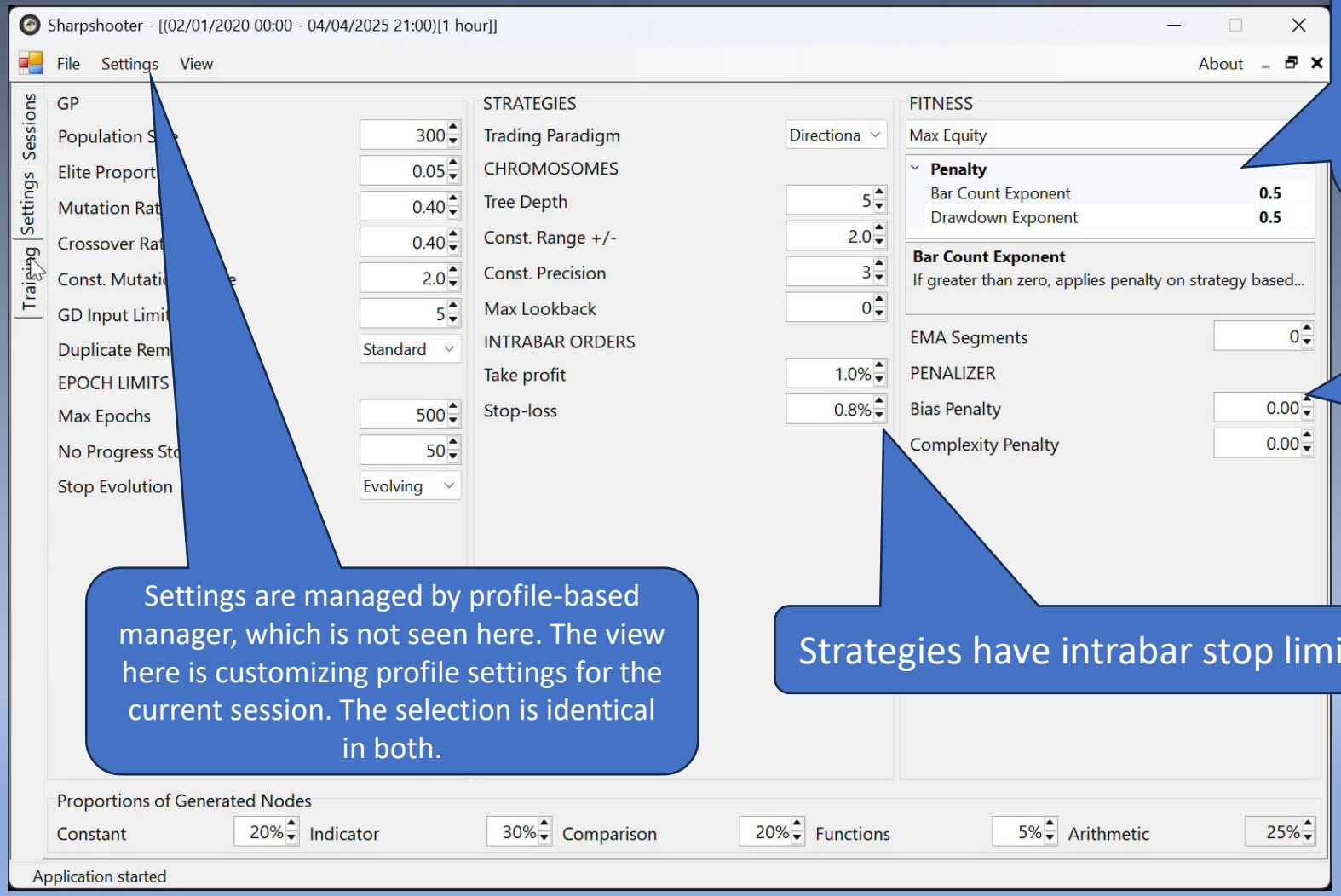


As most markets are not open and active 24/7, each instrument is assigned a trading schedule. This distinguishes missing data points and market being closed.

Indicator data can be either imported from external source, or calculated by the software. Sharpshooter has some built-in indicators, but the SDK allows defining proprietary indicators with the full power of .NET programming languges.

Sharpshooter has built-in market data storage. Data on events such as economic data and monetary policy announcements can also be added.

Tour of Sharpshooter — Settings



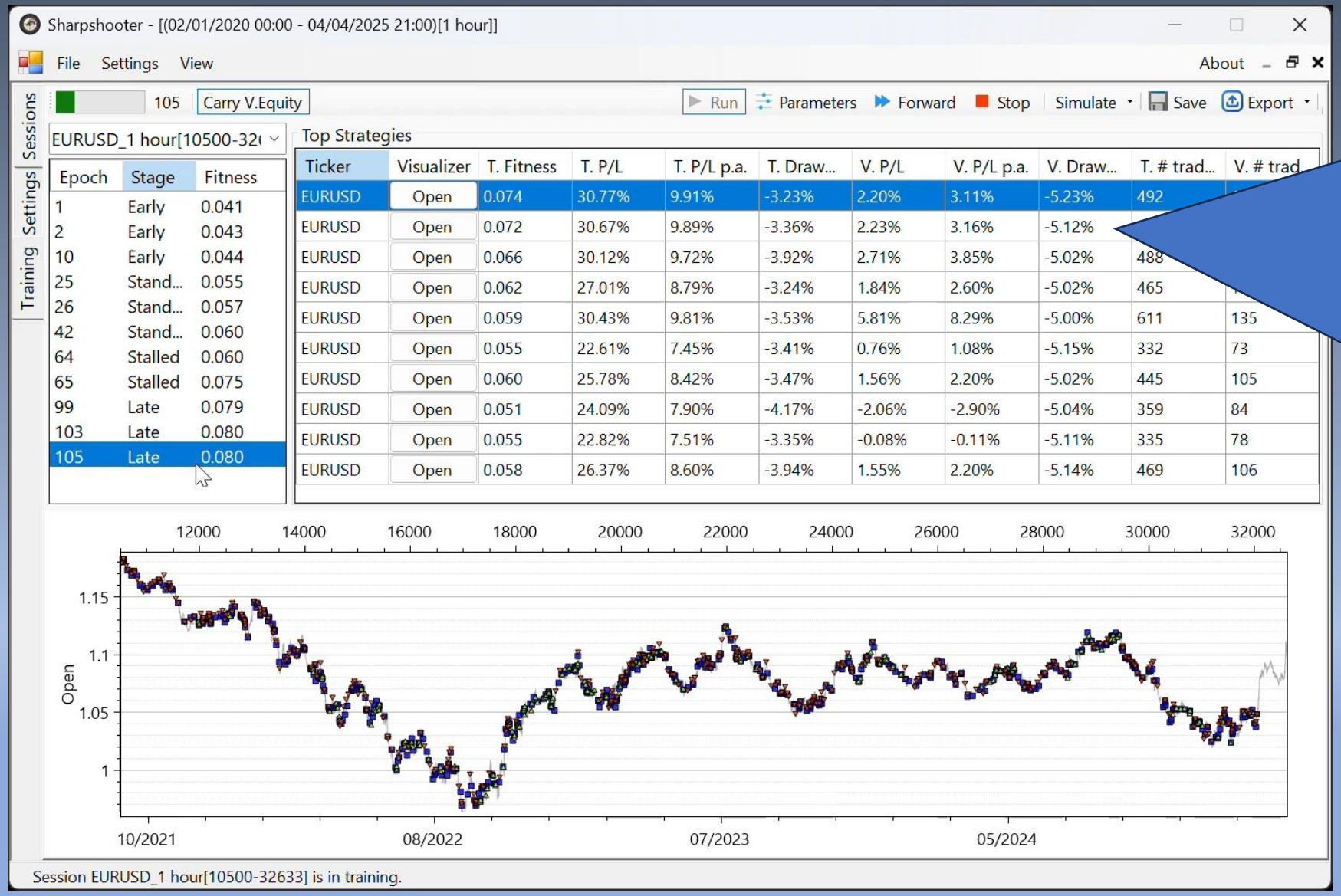
Many of the built-in fitness functions have tunable parameters. The standard function works well when tuned empirically to suit the particular instrument. This is essential for obtaining good results.

Bias penalty should be used at least when the training data that has overall trend. If possible, avoid such data sets altogether.

Settings are managed by profile-based manager, which is not seen here. The view here is customizing profile settings for the current session. The selection is identical in both.

Strategies have intrabar stop limits.

Tour of Sharpshooter — Finding Strategies



"T" refers to in-sample training and "V" to out-sample validation results. When training data extends to the most recent data, validation represents realizable P/L. It is therefore essential that strategies are consistently profitable in validation, across optimization sessions covering different time periods of the instrument.

Tour of Sharpshooter — Programmatic Access

The programmatic access is an enterprise feature that enables use of Sharpshooter without its graphical user interface. The typical use cases are:

- Explore a large set of instruments to identify which are the most promising ones.
- Update existing strategies frequently with the latest market data.

Tour of Sharpshooter — SDK

Enables extending Sharpshooter with proprietary

- Indicators. “Indicator” is understood here as any input data to the strategy. These are also available to the automatic strategy execution agent.
- Fitness functions, according to which Sharpshooter ranks quality of the strategies.

Strategy Execution Agent “SharpTrader”

Sharpshooter strategies are potentially challenging (or plainly impossible) to implement on commercially available algo trading software. This is positive, as it suggests having moat[1].

SharpTrader is the automatic strategy execution agent, into which strategies developed by Sharpshooter can be loaded from strategy files:

- Base software operates against Interactive Brokers low-level API, which also provides it with market data.
- Designed to be extensible to support other trading APIs.
- Supports trading “proxy instruments”, e.g., FX futures in place of the spot market, and other advanced features.

[1] Top market participants tend to use fully in-house solutions.

The Curse of Overfitting

All machine learning methods have the tendency to overfit, i.e., find "pseudopatterns" in random[1] variations of the particular training data.

- With recent market data (2020s), an issue also with Sharpshooter without tuning the optimization settings such fitness function. Readily apparent from lack of correlation between training and validation performance of strategies.
- Unique approach of fine-tunable fitness function delivers great improvement: around 80% of top strategies (per training data) for major FX pairs trade profitably post-training. Optimal parameters are instrument-dependent but sufficiently time-stable.

Old data from 2010s easy to "crack" even without careful tuning (market has clearly evolved to be more challenging).

[1] "Random" means here any behavior that does not have a repeating pattern. In many cases such may well have a rational origin.

Conclusion

Sharpshooter is a framework as well as ready for use software.

- Current testing suggest that the standard software can overcome the curse of overfitting to the extent that strategies are tradeable. Time-stability of these strategies open question.
- Serious institutional users would likely want to extend the software with substantial number of their own proprietary additions (framework approach).